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BCA (Sem-V) – Micro-
processor (503) DSE-I

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

Compulsory

1. Choose the correct alternative for each of the
following questions :

$2 \times 10 = 20$

(i) Which of the following is not a register in the 8085 microprocessor ?

- a) Program Counter (PC)
- b) Flag Register
- c) Accumulator (ACC)
- d) Command Register (CR)

(ii) Which of the following is true about the data bus ?

- a) It is 32 bits in length
- b) It is a bidirectional bus
- c) It is 16 bits in length
- d) Lower address bus lines (AD_0-AD_7) are called "Line number"

(iii) Which of the following interrupt lowest priority ?

- a) TRAP
- b) INTR

c) RST6.5

d) None of these

(iv) Which of the following instructions is used to load the HL register pair with an immediate 16-bit data in the 8085 microprocessor ?

- ~~a) LXI H, data~~
- b) LHLD addr
- c) SHLD addr
- d) LDAX addr

(v) Which of the following is a general-purpose register of microprocessor ?

- a) Program counter
- b) Instruction register
- c) Accumulator
- d) B,C,D,E

(vi) What does the instruction "CMA" do in the 8085 microprocessor ?

- a) Clears the accumulator
- b) Complements the contents of the accumulator
- c) Copies the contents of the accumulator to the memory
- d) Converts the contents of the accumulator to ASCII

(vii) Which of the following is not a status flag in microprocessor ?

- a) Zero flag
- b) Sign flag
- c) Parity flag
- d) Index flag

(viii) Which of the following instructions is used to load an immediate 8-bit data into

the accumulator in the 8085 microprocessor ?

- a) LDA
- b) STA
- ✓ c) MVI
- d) MOV

(ix) Which flag in the 8085 microprocessor indicates that the result of an operation is negative ?

- ✓ a) Sign flag (S)
- b) Zero flag (Z)
- c) Carry flag (CY)
- d) Parity flag (P)

(x) What is the function of the instruction "HLT" in the 8085 microprocessor ?

- ✓ a) Halts the microprocessor
- ✓ b) Hides the lower nibble of the accumulator

- c) Halves the contents of the accumulator
- d) Halts the timer

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

- 2. What are address bus and data bus?
- 3. What is immediate addressing mode? Give an example.
- 4. Write assembly language instructions for the following :
Move the content of memory location FC50 H to register C. The content of the memory Location FC50 H is 08.
- 5. List the different types of interrupts in descending order of priority.

S219/4

S219/4/3

(6)

Contd.

- 6. Write the pin description of the following two pins of 8085 microprocessor:

- (a) READY
- (b) HOLD

- 7. Write the short note on communication interfaces.

Section-C

Long Answer type Questions

Answer any **two** questions of the following :

15×2=30

- 8. Explain the types of registers with suitable figure.
- 9. Write an assembly language program to Subtract 32H from 49H.
The first number 49H is in the memory location 2501H.

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(7)

(Turn over)

The first number 32H is in the memory location 2502H.

The result is to be stored in the memory location 2503H.

10. Explain DMA controller with suitable figure.
11. Explain the different types of instruction formats and provide an example for each type.

↓
- Data transfer - COPY
↓
- Arithmetic - Add, Subtract, Increment/Decrement
→ logical - AND, OR, E-OR, Rotate, Compare
→ machine-control
→ Halt, Restart, interrupt